

SSGMCE	SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGG., SHEGAON - 444 203.		LABORATORY MANUAL	
	PRACTICAL EXPERIMENT INSTRUCTION SHEET			
	EXPERIMENT TITLE:	TO MANIPULATE & PROCESS DATA IN R.		
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SUBJECT: BIG DATA ANALYTICS			SEMESTER: VI	PAGE: 1 OF 4

Aim: To manipulate & process data in R.

Facilities: Computer with RStudio installed.

Theory:

Data Frames:

Data Frames are data displayed in a format as a table. Data Frames can have different types of data inside it, but each column should have the same type of data.

Use the `data.frame()` function to create a data frame:

Ex:- `emp.data <- data.frame(`

`emp_id = c(1:5),`

`emp_name = c("A","B","C","D","E"),`

`salary = c(100,500,800,90,700)`

`)`

`print(emp.data)`

Get the Structure of the Data Frame:

The structure of the data frame can be seen by using `str()` function.

Ex:- `str(emp.data)`

Access Items:

We can use single brackets `[ ]`, double brackets `[ [ ] ]` or `$` to access columns from a data

frame:

Ex:- `print(emp.data[2])`

`print(emp.data["emp_name"])`

`print(emp.data[[1]])`

`print(emp.data[["emp_id"]])`

`print(emp.data$salary)`

Add Column:

Ex:- `emp.data$dept <- c("IT","CSE","IT","ELPO","EXTC")`

`print(emp.data)`

Information of Rows and Columns:

Use the **dim()** function to find the amount of rows and columns in a Data Frame:

Ex:- `dim(emp.data)`

We can also use the **ncol()** function to find the number of columns and **nrow()** to find the number of rows:

Ex:- `ncol(emp.data)`

`length(emp.data)`

`nrow(emp.data)`

Use the **names()** function to view the names of the columns:

Ex:- `names(emp.data)`

Use the **rownames()** function to get the name of each row in the first column

Ex:- `rownames(emp.data)`

Combining Data Frames:

Use the **rbind()** function to combine two or more data frames in R vertically:

Ex:- `emp.data2 <- data.frame(`

```
emp_id = c(6:8),
emp_name = c("F","G","H"),
salary = c(900,500,400)
dept = c("IT","CSE","MECH")
)
print(emp.data2)
emp.newdata<-rbind(emp.data,emp.data2)
print(emp.newdata)
```

And use the **cbind()** function to combine two or more data frames in R horizontally:

```
Ex:- emp.data3 <- data.frame(
  CN1 = c(1:8),
  CN2 = c("W","X","Y","Z","W","X","Y","Z")
)
print(emp.data3)
emp.morecolumn<-cbind(emp.newdata,emp.data3)
print(emp.morecolumn)
```

Sort Variable Values:

To sort the values, use the **sort()** function:

```
Ex:- print(emp.morecolumn$salary)
      sort(emp.morecolumn$salary)
```

Max/Min:

```
Ex:- max(emp.morecolumn$salary)
      min(emp.morecolumn$salary)
```

We can use the **which.max()** and **which.min()** functions to find the index position of the max and min value in the table:

Ex:- `which.max(emp.morecolumn$salary)`

`which.min(emp.morecolumn$salary)`

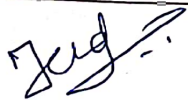
Or combine `which.max()` and `which.min()` with the `rownames()` function to get the name of the row

Ex:- `rownames(emp.morecolumn)[which.max(emp.morecolumn$salary)]`

`rownames(emp.morecolumn)[which.min(emp.morecolumn$salary)]`

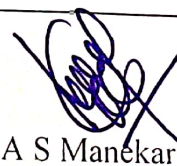
Conclusion: We have successfully manipulated and processed data in R.

Prepared By:



Prof. F I Khandwani

Approved By:



Prof. A S Maneekar